

Electrical Machine (1)
Essay Question Mid T2.2023-2024

Q1: Answer the following question ?

Q4. The magnetic circuit shown in Figure 2 has an infinitely permeable magnetic core. The parameters of the magnetic circuit are:

$g_1 = 5 \text{ mm}$	$A_1 = 5 \text{ cm}^2$	$N_1 = 80 \text{ turns}$
$g_2 = 5 \text{ mm}$	$A_2 = 5 \text{ cm}^2$	$N_2 = 100 \text{ turns}$
$g_3 = 10 \text{ mm}$	$A_3 = 10 \text{ cm}^2$	$N_3 = 125 \text{ turns}$

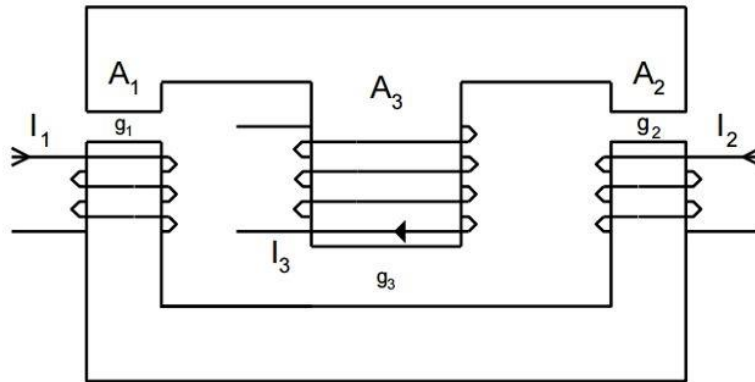


FIGURE 2

- a) A current of 12 A flows in the first coil, N_1 . The other coils are unexcited. Determine the flux and flux densities in each of the air gaps g_1 , g_2 and g_3 .

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Solution :

Given data:

$$g_1 = 5\text{mm}$$

$$A_1 = 5\text{cm}^2$$

$$N_1 = 80 \text{ turns}$$

$$g_2 = 5\text{mm}$$

$$A_2 = 5\text{cm}^2$$

$$N_2 = 100 \text{ turns}$$

$$g_3 = 10\text{mm}$$

$$A_3 = 5\text{cm}^2$$

$$N_3 = 125 \text{ turns}$$

$$I_1 = 12\text{A}$$

$$I_2 = 10\text{A}$$

$$I_3 = 8\text{A}$$

Formulae used:

$$1) \phi = \frac{\mu_0 N I A}{l}$$

$$2) B = \frac{\phi}{A} = \frac{\mu_0 N I}{l}$$

$$3) L = \frac{\phi}{I}$$

$$4) M = \frac{N \phi}{I}$$

where

ϕ = flux in weber

μ_0 = permeability of free space = $\frac{\mu_0}{4\pi}$

N = no. of turns

I = current through the coils in ampere

l = length of core in meter

B = magnetic flux density in weber / m^2

L = Inductance in henry

M = mutual inductance in henry

Now,

a) For flux densities through g_1 , g_2 and g_3 respectively :

$$B_1 = \frac{\phi}{A} = \frac{\mu_0 N_1 I_1}{l} = \frac{\mu_0 N I}{g_1} = \frac{4\pi \times 10^{-7} \times 80 \times 12}{5 \times 10^{-3}} = 0.2413 \text{Wb} / \text{m}^2$$

$\therefore$, second and third coils are unexcited, i.e., $I_2 = I_3 = 0\text{A}$

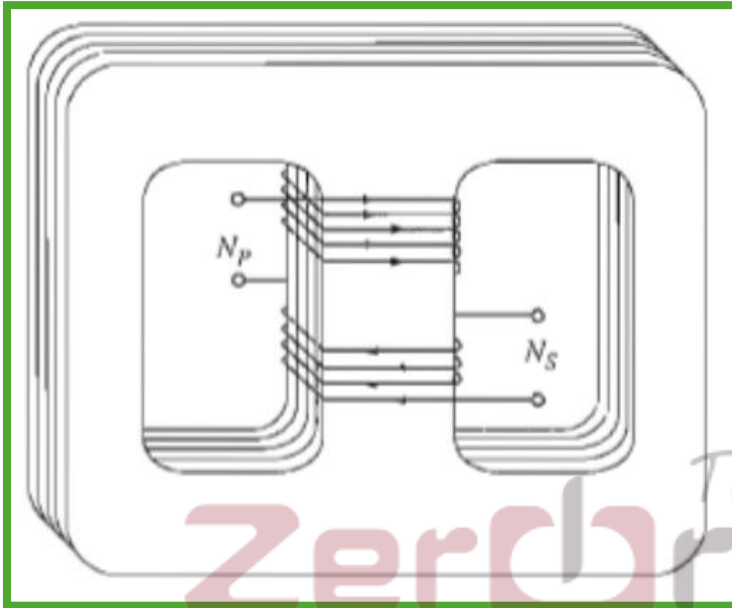
$\Rightarrow$ flux densities through g_2 and g_3 will be $0\text{Wb} / \text{m}^2$

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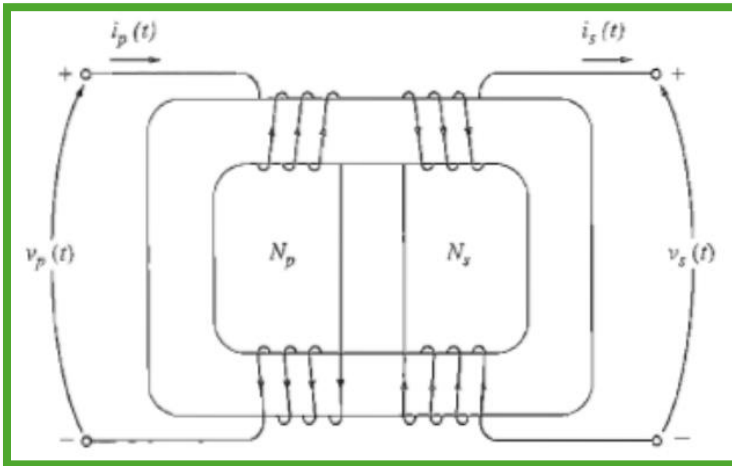
Q2: Mention and draw the types and construction of transformers ?

Solution :

1) Shell form :



2) Core form : *Computer & Networks Engineers*





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Choose the correct answer:

Q1)

- 1) The law describing the electromotive force induced in a conductor placed in a varying magnetic field is
 A) Ohm's law
 B) Gauss's law
 C) Faraday's law
 D) Ampere's law

- 2) The unit of the magnetic flux is:
 A) Henry
 B) Webber
 C) Tesla
 D) Ampere/meter

- 3) A constant current flows in a horizontal wire in the plane of the paper from east to west. The point at which the direction of magnetic field is from North to South is:
 A) Directly above the wire
 B) In the plane of the paper, in the north side of the wire
 C) Directly below the wire
 D) In the plane of the paper, in the south side of the wire

- 4) The electromotive force induced in a conductor of length l moving at velocity $\vec{v}$ in a magnetic field having flux density $\vec{B}$ is:
 A) $e_{ind} = (\vec{v} \times \vec{B}) \cdot \vec{l}$
 B) $e_{ind} = (\vec{v} \cdot \vec{B}) \cdot \vec{l}$
 C) $e_{ind} = (\vec{v} \cdot \vec{B}) \cdot \vec{l}$
 D) $e_{ind} = (\vec{v} \times \vec{B}) \cdot \vec{l}$

- 5) The force F on a current carrying conductor of length l and current I , placed perpendicular in a uniform magnetic field of flux density B is:
 A) $F = \frac{l}{B}$
 B) $F = \frac{B}{l}$
 C) $F = Bl$
 D) $F = \frac{l}{Bl}$

Q2)

- 6) In the power systems, the auto transformer is preferred over the ordinary two winding transformer due to:
 A) Safety of operation
 B) Availability of taps
 C) Material saving
 D) All the mentioned

- 7) The full-load copper loss of a transformer is 2000W. At half-load, the copper loss will be:
 A) 400W
 B) 2000W
 C) 1000W
 D) 500W

- 8) A transformer with primary winding having 100 turns, secondary winding having 500 turns, if 200 V is applied to the primary winding, the voltage at the secondary winding will be:
 A) 2000V
 B) 1000V
 C) 500V
 D) 100V

- 9) A transformer with primary winding having 100 turns, secondary winding having 500 turns, if 200 A is flowing into the primary winding, the current out of the secondary winding will be:
 A) 40A
 B) 100A
 C) 500A
 D) 1000A

- 10) The winding of the transformer that has less cross-sectional area of the conductor is:
 A) primary winding
 B) secondary winding
 C) high voltage winding
 D) low voltage winding

Q3)

- 11) If the rated speed of a DC motor is 1200RPM and the rated torque is 40Nm, the rated power of this motor is:
 A) 30kW
 B) 5kW
 C) 3kW
 D) 0.5kW



12) If the work done by a motor is $1000J$ while its torque is $10N.m$, the number of complete revolution done by the shaft of this motor is:

A) 100π

B) 50π

C) $\frac{50}{\pi}$

D) $\frac{100}{\pi}$

13) The commutator pitch for the simplex retrogressive winding:

C) $+1$

D) $+2$

14) If the field winding of a lightly loaded DC shunt motor, gets opened, the:

B) motor speed will decrease slightly

C) motor will pickup very high speed.

15) If a DC motor have a full-load speed of 750 RPM and speed-regulation of 10% , the no-load speed of this motor is:

D) 675 RPM

C) 700 RPM

B) 800 RPM

A) 825 RPM

16) In the separately excited DC motor, the relation between the motor speed, n , and the field current, I_f is:

A) n is directly proportional to I_f

C) n is inversely proportional to I_f

B) n is directly proportional to the square of I_f

D) n is inversely proportional to the square of I_f

17) In the electric generator, the energy conversion is:

A) from mechanical to electrical

C) from electrical to mechanical

B) from mechanical to electrical

D) from electrical to electrical

18) In the DC series machine, the magnetic flux is proportional to the:

A) square of the armature current

C) square root of the armature current

B) armature current

D) cube of the armature current

19) The DC motor that is most suitable for producing high starting torque is the:

A) separately-excited

B) shunt excited

C) serially excited

D) differentially compound

20) If the load of the shunt DC motor is reduced, its speed will:

A) increase abruptly

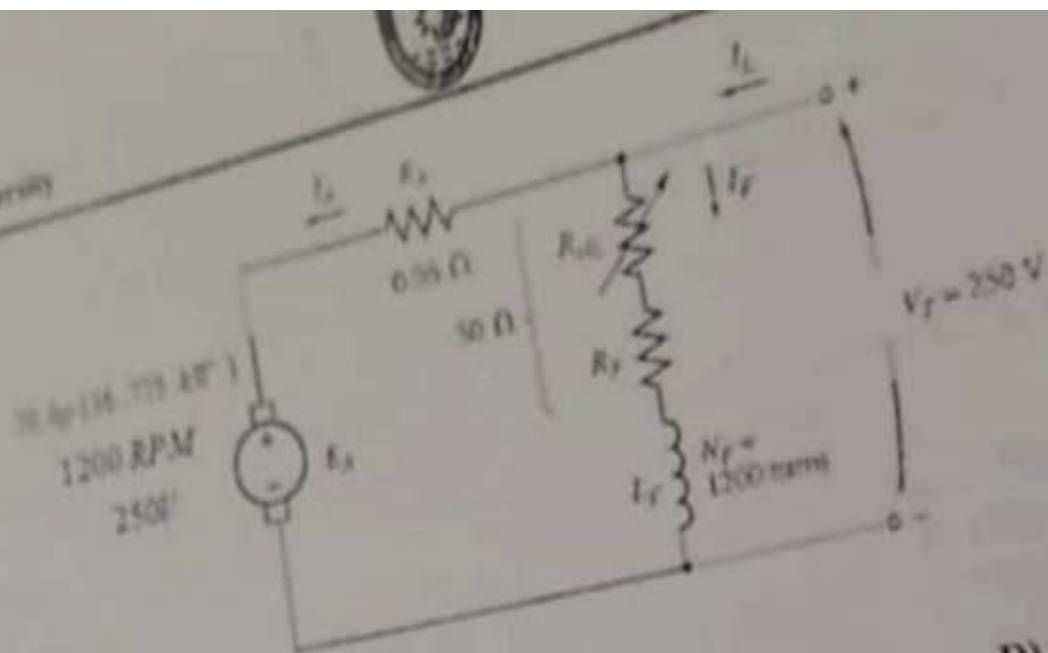
C) decrease in proportion to speed reduction

B) increase in proportion to speed reduction

D) remain almost constant

Q4) Solve showing your work:

The figure below shows the diagram of a $50\text{hp}(36.775\text{kW})$, 250V , 1200RPM dc shunt motor with compensating windings and an armature resistance of 0.06Ω . The field circuit of the motor has a total resistance ($R_{se} + R_f = 50\Omega$) which produces a no-load speed of 1200RPM . There are 1200 turns per pole on the shunt field winding.



- 21) The rated current of this motor is:
 A) 147.1A
 B) 199.5A
 C) 248.6A
 D) 575.5A
- 22) The rated torque of this motor is:
 A) 585.3 N.m
 B) 379.5N.m
 C) 292.7 N.m
 D) 146.3N.m
- 23) The speed of this motor when the line current is 100A:
 A) 987 RPM
 B) 1020 RPM
 C) 1157.6 RPM
 D) 1173 RPM
- 24) The speed of this motor at rated current is:
 A) 987 RPM
 B) 1020 RPM
 C) 1157.6 RPM
 D) 1173 RPM
- 25) The speed regulation of this motor is:
 A) 2.78%
 B) 3.66%
 C) 5.67%
 D) 6.75%

1	2	3	4	5	6	7	8	9	10	11	12
C	B	D	B	C	D	C	A	C	A	A	A
13	14	15	16	17	18	19	20	21	22	23	24
D	A	C	B	B	A	C	A	C	B	B	B
							A				

For autotransformer

a. What are the uses of an autotransformer?

- Widely used in synchronous and induction motors.
- Easy to find application in AC feeders due to its help in increasing the voltage level.
- It is used in almost all the electrical appliances.
- It has been used for so long in power distribution and transmission system.
- Railway systems

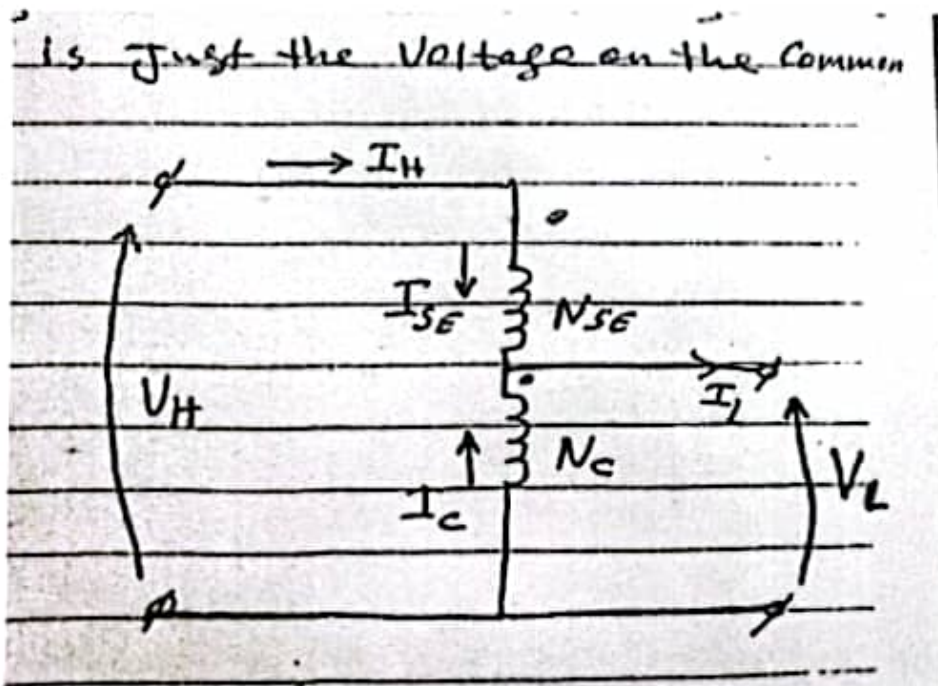
b. What is the difference between a power transformer and an autotransformer?

small rises may be made necessary

The principal disadvantage of autotransformers is that, unlike ordinary transformers, *there is a direct physical connection between the primary and the secondary circuits*, so the *electrical isolation* of the two sides is lost. If a particular application does not require electrical isolation, then the autotransformer is a convenient and *inexpensive* way to tie nearly equal voltages together.

c. Draw the step-down autotransformer circuit

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d. Find the relationship between low current and high current

Sides of the Tr-r can be found by noting that:-

$$I_L = I_c + I_{sc}$$

from equation $N_c I_c = N_{sc} I_{sc}$

$$I_c = \frac{N_{sc}}{N_c} I_{sc}$$

so
$$I_L = \frac{N_{sc}}{N_c} I_{sc} + I_{sc}$$

Finally noting that $I_H = I_{sc}$

$$I_L = \frac{N_{sc}}{N_c} I_H + I_H$$

$$I_L = \frac{N_{sc} + N_c}{N_c} I_H$$

$$\boxed{\frac{I_L}{I_H} = \frac{N_{sc} + N_c}{N_c}}$$

3) Solve and show your work

(9 Marks)

A single-phase transformer has 100 turns on the secondary and 1000 turns on primary. If transformer is connected two 110 V supply while 3 kVA load connected to the secondary winding. Assuming ideal transformer, find.

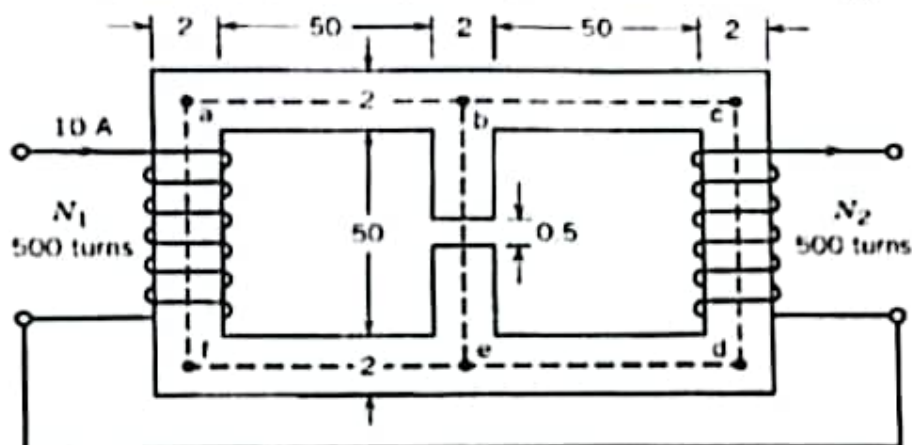
- A load impedance
- B load voltage
- C load impedance as referred to the primary

a) $V_2 = \frac{N_2}{N_1} V_1$
 $= \frac{100}{1000} 110 = 11 \text{ V}$

b) $I_2 = \frac{P}{V} = \frac{3000}{11} = 272.72 \text{ A}$
 $Z_2 = \frac{11 \text{ V}}{272.72 \text{ A}} = 0.04 \Omega$

c) $a = \frac{1000}{100} = 10$
 $Z' = a^2(Z)$
 $(10)^2(0.04) = 4 \Omega$

A magnetic circuit as shown below with relative permeability of the ferromagnetic material is 1200. Neglect magnetic leakage and fringing. All dimensions are symmetric in centimeters and magnetic material has square cross section area. If the length for b-e (only) is 51.5 cm, Find the airgap flux, airgap flux density, and magnetic field intensity in the air gap.



Solution

The mean magnetic paths of the fluxes are shown by dashed lines in Fig. E1.3a. The equivalent magnetic circuit is shown in Fig. E1.3b.

$$F_1 = N_1 I_1 = 500 \times 10 = 5000 \text{ At}$$

$$F_2 = N_2 I_2 = 500 \times 10 = 5000 \text{ At}$$

$$\mu_r = 1200 \mu_0 = 1200 \times 4\pi \times 10^{-7}$$

$$R_{\text{bale}} = \frac{l_{\text{bale}}}{\mu_r A_c}$$

$$= \frac{3 \times 52 \times 10^{-2}}{1200 \times 4\pi \times 10^{-7} \times 4 \times 10^{-4}}$$

$$= 2.58 \times 10^6 \text{ At/Wb}$$

$$\mathcal{R}_{\text{bdc}} = \mathcal{R}_{\text{baf}}$$

$$\mathcal{R}_g = \frac{l_g}{\mu_0 A_g}$$

$$= \frac{5 \times 10^{-3}}{4\pi 10^{-7} \times 2 \times 2 \times 10^{-4}}$$

$$= 9.94 \times 10^6 \text{ At/Wb}$$

$$\mathcal{R}_{\text{bc(core)}} = \frac{l_{\text{bc(core)}}}{\mu_c A_c}$$

$$= \frac{51.5 \times 10^{-2}}{1200 \times 4\pi 10^{-7} \times 4 \times 10^{-4}}$$

$$= 0.82 \times 10^6 \text{ At/Wb}$$

The loop equations are

$$\Phi_1(\mathcal{R}_{\text{baf}} + \mathcal{R}_{\text{bc}} + \mathcal{R}_g) + \Phi_2(\mathcal{R}_{\text{bc}} + \mathcal{R}_g) = F_1$$

$$\Phi_1(\mathcal{R}_{\text{bc}} + \mathcal{R}_g) + \Phi_2(\mathcal{R}_{\text{bdc}} + \mathcal{R}_{\text{bc}} + \mathcal{R}_g) = F_2$$

$$\Phi_1(13.34 \times 10^6) + \Phi_2(10.76 \times 10^6) = 5000$$

$$\Phi_1(10.76 \times 10^6) + \Phi_2(13.34 \times 10^6) = 5000$$

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The air gap flux density is

$$B_g = \frac{\Phi_g}{A_g} = \frac{4.134 \times 10^{-4}}{4 \times 10^{-4}} = 1.034 \text{ T}$$

The magnetic intensity in the air gap is

$$H_g = \frac{B_g}{\mu_0} = \frac{1.034}{4\pi 10^{-7}} = 0.822 \times 10^6 \text{ At/m}$$